

CS 630: Software Testing and Quality Assurance

9-1 Project Two: Designing a Test and Quality Strategy

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Test Strategy and Approach: Amazon Path-to-Purchase

This test strategy uses the Amazon path-to-purchase as the public reference feature: product selection, cart management, checkout, payment, and order confirmation. The purpose is not to test Amazon itself, but to demonstrate a disciplined strategy that SquareCalc could adapt while rescuing a high-risk customer-facing financial website. SquareCalc currently has no formal test process, no documented user-journey or user-acceptance tests and relies heavily on developers testing their own work during hardening periods. Its shared test environment can also contain mismatched service versions, while test data is inconsistent and may be stale (SquareCalc Development Process Overview, n.d.). The proposed approach therefore shifts testing earlier, assigns explicit ownership, automates repeatable checks in continuous integration, protects payment and personal data, and establishes release gates based on evidence rather than the end of a sprint. The strategy prioritizes failures that could prevent a purchase, expose sensitive data, charge a customer incorrectly, or cause abandonment. Testing will combine risk-based functional testing, API and integration testing, security testing, accessibility and usability evaluation, performance testing, regression automation, exploratory testing, and structured user acceptance testing (UAT). Success will be measured through pass rates, critical defect escape rate, requirements/risk coverage, defect removal effectiveness, checkout completion reliability, and mean time to verify fixes. These measures give product owners and management a concise view of release risk while giving engineering teams actionable information during each sprint.

Selected Feature and Risk Analysis

The selected feature is the end-to-end path-to-purchase on Amazon, beginning when a customer selects an item and ending when the system displays a confirmed order. This slice is appropriate because it crosses the web interface, identity/session handling, inventory and pricing logic, APIs, payment processing, and confirmation services. It also resembles the type of high-consequence transaction flow SquareCalc must be prepared to protect in a personal-finance product, particularly because SquareCalc states that it must comply with PCI and Sarbanes-Oxley requirements and is audited annually (SquareCalc Development Process Overview, n.d.).

Functional risks include incorrect quantities or prices in the cart, invalid discounts or taxes, stale inventory, duplicate orders, payment authorization without a completed order, failed confirmation messages, and loss of cart state between devices or sessions. Integration risk is especially important because a checkout journey depends on multiple services. SquareCalc already reports that test dependencies may be unavailable or mis-versioned in shared environments, so an apparently isolated defect may be a contract or environment problem.

Security risks include account takeover, session hijacking, insecure direct object references, injection, cross-site scripting, cross-site request forgery, exposure of payment or personally identifiable information, weak authorization around saved addresses or payment methods, and sensitive data leaking into logs. The payment step also creates compliance risk if cardholder data is stored, transmitted, logged, or retained outside approved controls. Abuse cases such as repeated coupon attempts, bot-driven checkout, and replayed payment requests must be considered because a function can work correctly for a normal user while remaining unsafe under adversarial use. User-experience risks include confusing error messages, inaccessible form of controls, keyboard traps, poor mobile responsiveness, unclear shipping or total-cost information, lost progress after an error, and slow page or API response. These failures may not

crash into the system, but they can reduce trust and conversion. Accessibility is treated as a quality requirement rather than a final visual check; checkout must remain understandable and operable with keyboard navigation, screen readers, zoom, and common mobile form behaviors.

Mitigation Plan: Strategy, Tools, Methods, and Roles

Testing will follow a risk-based, layered model. During refinement, the product owner, developer, and QA lead will convert acceptance criteria into testable examples and identify high-risk data and service dependencies. Developers will maintain unit and component tests. API tests will validate contracts, status codes, authorization, idempotency, price calculations, and negative conditions before full browser tests are executed. Automated end-to-end tests will cover a small set of critical purchase journeys, while exploratory testing will target edge cases that scripted tests may miss. Security testing will include threat-based negative tests and dependency/static scanning; performance testing will validate checkout behavior under expected and peak load. UAT will use documented scenarios and named business approvers rather than informal testing during hardening. Recommended tools are Jira for requirements, defects, traceability, and dashboards; the existing CI server for automated quality gates; Playwright or Selenium for cross-browser end-to-end automation; Postman/Newman or an equivalent API framework for service tests; OWASP ZAP for dynamic security scanning; a software composition analysis tool for vulnerable dependencies; and JMeter or k6 for load testing. Accessibility checks can begin with axe-core automation and be supplemented by keyboard and screen-reader evaluation. Tool choice is secondary to integration: results should be published automatically to the build and linked to the Jira story or defect so that evidence is visible to the entire delivery team.

Test data will be versioned and intentionally designed. Known-good synthetic users, carts, addresses, inventory states, coupons, and payment tokens will be created for positive, boundary, and negative scenarios. Production financial data will not be copied into routine test automation. This directly addresses SquareCalc's inconsistent practice in which some teams use CSV fixtures while others use aging copies of production data with personal information removed. Service virtualization or mocks will be used for deterministic tests when a dependent service is unstable; a smaller set of integration tests will still run against real test/stage services to detect contract and deployment problems. Roles are explicit. The product owner owns acceptance criteria and business priority. Developers own unit/component quality and correction of defects. A QA lead owns the test strategy, risk register, traceability, exploratory charters, UAT coordination, and release-quality recommendation. DevOps/platform engineering owns reliable environments, test execution in CI, observability, and test-data provisioning mechanisms. Security specialists review threat scenarios and high-risk findings. UX/design supports accessibility and usability validation. Management resolves risk acceptance when exit criteria are not met. This shared model avoids making QA the sole owner of quality while also correcting SquareCalc's current problem that no one organizes hardening-week testing.

Measurable Test Goals

- Achieve 100% automated execution of the defined critical purchase smoke suite on every release candidate, with 100% pass required before promotion.
- Maintain at least 95% pass rate for the full planned regression suite, with every failure triaged and no unresolved Severity 1 or Severity 2 defect at release.

- Trace 100% of high-risk requirements and identify critical risks to at least one test case and recorded result.
- Keep critical production defect escape rate below 2% of total critical/high defects identified for a release and target zero defects involving unauthorized data access or duplicate/incorrect charges.
- Verify corrected Severity 1 defects within four business hours and Severity 2 defects within one business day after a deployable fix is available.
- Meet agreed performance thresholds for the checkout journey under expected and peak load, with no material increase in error rate and no unhandled transaction loss.

Test Metrics and Rationale

Test pass rate will show whether the release candidate satisfies the planned automated and manual checks, but it will never be interpreted alone because a high pass rate is meaningless if high-risk behavior is untested. Risk/requirements of coverage is therefore paired with pass rate to show whether critical business and security scenarios have evidence. Critical defect escape rate measures the effectiveness of preproduction testing by identifying serious defects first discovered after release. Defect removal effectiveness compares defects removed before release with the total found before and after release, helping management see whether quality controls are catching problems early. Defect aging and mean time to verify fixes to measure responsiveness and flow. Long-lived severe defects indicate either ownership problems, unstable environments, or insufficient engineering capacity. Automated regression stability will track flaky-test rate because unreliable automation creates false confidence and wastes time. Checkout completion reliability and transaction error rate connect technical testing to the business

outcome: a release is not high quality if tests pass while legitimate users cannot complete the journey. Security findings will be tracked by severity, age, and recurrence, and accessibility findings will be tracked against critical user journeys. Together, these metrics balance test execution, risk coverage, product behavior, and improvement over time rather than rewarding the team for simply running more tests.

Entry, Exit, and Stakeholder Alignment

A release candidate enters formal system testing only when the build is deployable; unit/component tests pass, required services and synthetic data are available, and acceptance criteria are testable. Exit requires the critical smoke suite to pass, no open Severity 1 or Severity 2 defects unless management formally accepts a documented risk, high-risk requirements to have traceable evidence, security findings above the agreed threshold to be resolved, and UAT approval for the purchase journey. At each sprint demo, the QA lead will present a short quality dashboard: critical risks, coverage, open severe defects, escaped defects, automation health, and release recommendation. This aligns testing with SquareCalc's existing sprint demos, retrospectives, CI practices, Jira tracking, and release trains instead of creating a separate process that teams are likely to bypass.

Appendix A: Sample Test Execution Evidence

This appendix demonstrates the evidence standard recommended for SquareCalc. A limited public-site execution was performed on August 31, 2026, using Google Chrome on a Mac desktop. Each result records the visible URL, timestamp, execution context, reproduction procedure, expected and actual result, status, and screenshot reference. Browser version and

exact pixel viewport were not captured during this sample run; under the proposed SquareCalc process, both would be mandatory evidence fields and collected automatically where possible.

Test / Evidence	Procedure	Expected vs. Actual	Result
TC-01 / Fig. 1 12:36:38 PM	Open Amazon eero Pro 6E product page. Confirm product, price, stock, quantity, and Add to Cart. Visible URL: amazon.com/Amazon-eero-Wi-Fi-router-newest/dp/B091G65HH6/...	Expected: purchasable product with visible price/cart control. Actual: \$179.99, In Stock, quantity 1, Add to cart visible.	PASS
TC-02 / Fig. 2 12:37:36 PM	Select Add to cart and open Shopping Cart. Visible URL: amazon.com/gp/cart/view.html?ref_=nav_cart	Expected: one selected product; subtotal \$179.99. Actual: quantity 1; subtotal \$179.99.	PASS
TC-03 / Fig. 3 12:38:53 PM	Select + once to change quantity 1 → 2; wait for recalculation.	Expected: $2 \times \$179.99 = \359.98 . Actual: quantity 2; subtotal \$359.98.	PASS
TC-04 / Fig. 4 12:39:40 PM	Select – once to return quantity 2 → 1; wait for recalculation.	Expected: quantity 1; subtotal \$179.99. Actual: quantity 1; subtotal \$179.99.	PASS

Scope limitation: This sample does not claim comprehensive validation of Amazon or payment processing. No purchase, sign-in, address entry, or payment data entry was performed. The evidence demonstrates traceability for representative functional risks and shows how a result can be reproduced and defended.



Amazon eero Pro 6E mesh wifi router - Supports internet plans up to 2.5 Gbps, Coverage up to 2,000 sq. ft., Connect 100+ devices, 1-pack

Visit the eero Store
4.3 (6,108)
1 sustainability feature
1K+ bought in past month

Limited time deal
-10% \$179⁹⁹
List Price: \$199.99

Get \$50 off instantly: Pay \$129.99 \$179.99 upon approval for Amazon Visa. No annual fee.

FREE Returns
Save 20% with Trade-In

There's a newer model of this product - learn more

\$179⁹⁹
FREE delivery Saturday, September 5

Or Prime members get FREE delivery Tomorrow, September 1. Order within 11 hrs 21 mins. Join Prime

Delivering to Des Plaines 60016 - Update location

In Stock

Quantity: 1

Add to cart

Buy Now

Shipper / Seller Amazon.com
Returns FREE 30-day refund/replacement
Support Product support included

See more
☐ This is a gift

Get a \$50 Gift Card that can be used towards this order upon approval for Amazon Visa



Shopping Cart

Price



Amazon eero Pro 6E mesh wifi router - Supports internet plans up to 2.5 Gbps,

In Stock

FREE delivery Sat, Sep 5 available at checkout

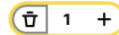
FREE Returns

Carbon impact

This is a gift Learn more

Configuration: 1-pack

Style: without eero Plus



Delete Save for later Compare with similar items Share

10% off

Limited time deal

\$179.99

List Price: \$199.99

Subtotal (1 item): \$179.99

Your order qualifies for FREE delivery. Choose this option at checkout.

Subtotal (1 item): \$179.99

This order contains a gift

Proceed to checkout

Related products with fast delivery



Amazon eero 7 dual-band mesh wifi router

★★★★★ 2,127

\$399.99

Get it as soon as Saturday, Sep 5

FREE Shipping by Amazon

Add to cart



Wall Mount Holder for eero

Delivering to Des Plaines 60016
[Update location](#)

All ▾

Search Amazon



EN ▾

Hello, sign in
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& Orders](#)[Health AI](#)[Amazon Haul](#)[Medical Care ▾](#)[Amazon Basics](#)[Best Sellers](#)[Prime ▾](#)[New Releases](#)[Groceries](#)[Whole Foods](#)[Books](#)[Today's Deals](#)[Get a \\$50 Gift Card](#) that can be used towards this order upon approval for **Amazon Visa**

Shopping Cart

Price

**Amazon eero Pro 6E mesh wifi router -
Supports internet plans up to 2.5 Gbps,****In Stock****FREE delivery Sat, Sep 5** available at checkout**FREE Returns** **Carbon impact** ▾☐ This is a gift [Learn more](#)**Configuration:** 1-pack**Style:** without eero Plus

- 2 +

[Delete](#)[Save for later](#)[Compare with similar items](#)[Share](#)**10% off**
Limited time deal
\$179.99

List Price: \$199.99

Subtotal (2 items): \$359.98 **Your order qualifies for FREE delivery.**
Choose this option at checkout.**Subtotal (2 items): \$359.98**☐ This order contains a gift[Proceed to checkout](#)

Related products with fast delivery

**Amazon eero 7 dual-...**
★★★★☆ 2,127**\$399⁹⁹**Get it as soon as **Saturday, Sep 5**

FREE Shipping by Amazon

[Add to cart](#)**Wall Mount Holder fo...**
★★★★☆ 539**\$24⁹⁸**Get it as soon as **Saturday,**

The price and availability of items at Amazon.com are subject to change. The Cart is a temporary place to store a list of your items and reflects each item's most recent price. [Learn more](#)

Get a [\\$50 Gift Card](#) that can be used towards this order upon approval for Amazon Visa



Shopping Cart

Price



Amazon eero Pro 6E mesh wifi router - Supports internet plans up to 2.5 Gbps,

In Stock

FREE delivery **Sat, Sep 5** available at checkout

FREE Returns

Carbon impact

☐ This is a gift [Learn more](#)

Configuration: 1-pack

Style: without eero Plus

1

Delete Save for later Compare with similar items Share

10% off
Limited time deal
\$179.99
List Price: \$199.99

Subtotal (1 item): **\$179.99**

✓ Your order qualifies for **FREE** delivery.
Choose this option at checkout.

Subtotal (1 item): **\$179.99**

☐ This order contains a gift

Proceed to checkout

Related products with fast delivery



Amazon eero 7 dual-band mesh wifi router

★★★★☆ 2,127

\$399.99

Get it as soon as **Saturday, Sep 5**

FREE Shipping by Amazon

Add to cart



Wall Mount Holder for eero

References:

SquareCalc Development Process Overview. (n.d.). CS 630 Project Two course supporting material. Southern New Hampshire University.

SquareCalc Company Overview. (n.d.). CS 630 Project Two course supporting material. Southern New Hampshire University.

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PCI Security Standards Council. (2024). Payment Card Industry Data Security Standard (PCI DSS), Version 4.0.1.